

Security measures taken for OpenForge

- **Standard practices:** NeGD follows the standard software development practices of uniform coding standards, guidelines and reviews. It complies with OWASP (Open Web Application Security Project) security standards and guidelines. Every product release is reviewed and tested internally for security vulnerabilities before it is deployed.
- **256-bit SSL encryption:** OpenForge uses 256-bit secure socket layer (SSL) encryption for information transmitted during any activity.
- **ISO 27001 certified data centre:** The application is hosted in an ISO 27001 security certified data centre.
- **Data redundancy:** Data is backed up in a secure environment with proper redundancy.
- **Security audit:** The application has been audited by a recognised third party audit agency for security and vulnerability, after which a security audit certificate has been obtained.

Multifarious efforts to get contributions

NeGD has planned some key initiatives to encourage various IT teams and the community to share their code on OpenForge. Nayak says that his team is looking at internal evangelism for state level contributions as well as developing

a systematic and constructive process to receive contributions from the community. "We want to promote a culture of treating the software code as an asset. This will need internal evangelism," Nayak told *Open Source For You*.

The team is also planning to add credibility to the project by featuring some of the high profile projects done by the government. "We will especially try to highlight work done by the bigger government departments and the well-known applications," Nayak adds.

For a successful open source based future

Alongside community contribution, the main focus of the platform is to introduce the philosophy of 'open, share and collaborate' to government departments.

"There is no dearth of young and enthusiastic talented people in academia and industry who are willing to contribute to government applications. Therefore, we sincerely hope that OpenForge will bridge this gap between e-governance and open source, bringing the community and the government together," Chauhan concludes. 

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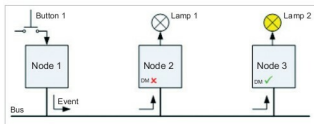


Figure 8: VSCP operations

VSCP (Very Simple Control Protocol): This is more a framework than a protocol. VSCP is highly scalable, has a low footprint and is a free-cum-open source solution for device discovery and identification, device configuration, autonomous device functionality and secure firmware updates. VSCP makes things interact at the application layer. It makes use of CAN, RS-232, Ethernet, TCP/IP, MQTT and 6LoWPan.

VSCP uses an event format and supports global unique identifiers for nodes, thus making a node identifiable no matter where it is installed in the world. Besides, it includes a register model in order to provide a flexible common interface for node configuration and a model for controlling the functionality of each node. VSCP does not make any assumptions regarding the lower level system used to realise the physical interconnection with the node; therefore, it works with different transport mechanisms such as Ethernet, TCP/IP,

wireless, Zigbee, Bluetooth, CAN, GPRS, RS-232 and USB.

VSCP is event-based. Every time an event occurs, it is broadcast to all other nodes on the network. From there, each node decides on its own if the event received needs to be processed or not. The final decision depends on the node's decision matrix, which is made up of a number of 'if condition> then action>' lines, where the condition> is evaluated based on the fields present in the VSCP datagram broadcast to the network. 

References

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